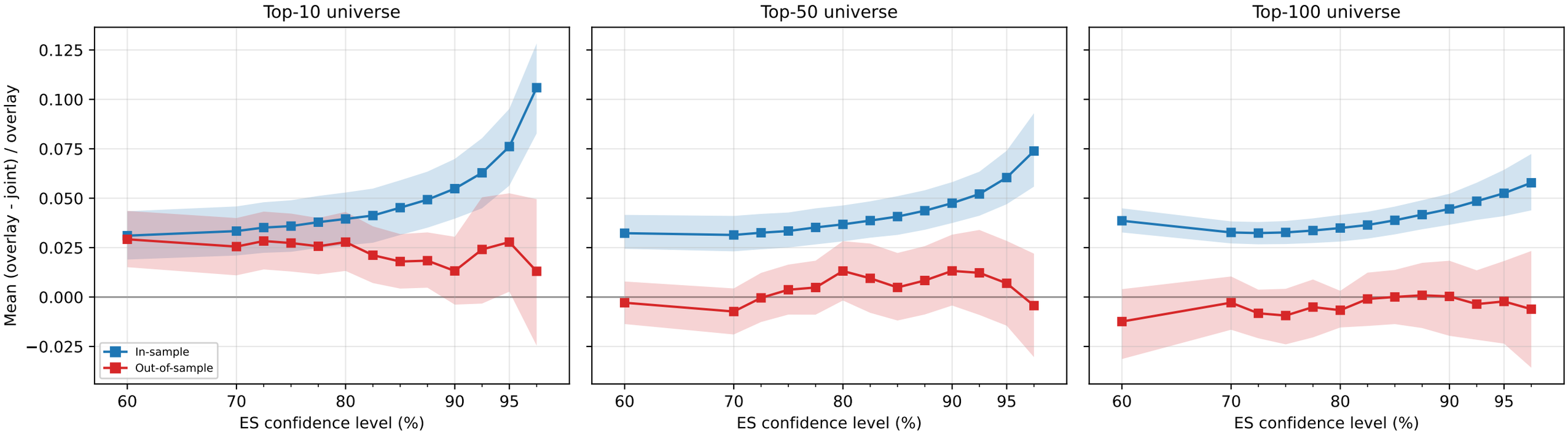
Joint vs overlay optimization across Expected Shortfall confidence levels
Top-10 universe
Top-50 universe
Top-100 universe
Mean (overlay - joint) / overlay
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
In-sample
Out-of-sample
60
70
80
90
95
ES confidence level (%)